

Deep learning approach to skin layers segmentation in inflammatory dermatoses

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ABSTRACT

Monitoring skin layers with medical imaging is critical to diagnosing and treating patients with chronic inflammatory skin diseases. The high-frequency ultrasound (HFUS) makes it possible to monitor skin condition in different dermatoses. Accurate and reliable segmentation of skin layers in patients with atopic dermatitis or psoriasis enables the assessment of the treatment effect by the layer thickness measurements. The epidermis and the subepidermal low echogenic band (SLEB) are the most important for further diagnosis since their appearance is an indicator of different skin problems. In medical practice, the analysis, including segmentation, is usually performed manually by the physician with all drawbacks of such an approach, e.g., extensive time consumption and lack of repeatability. Recently, HFUS becomes common in dermatological practice, yet it is barely supported by the development of automated analysis tools. To meet the need for skin layer segmentation and measurement, we developed an automated segmentation method of both epidermis and SLEB layers. It consists of a fuzzy c-means clustering-based preprocessing step followed by a U-shaped convolutional neural network. The network employs batch normalization layers adjusting and scaling the activation to make the segmentation more robust. The obtained segmentation results are verified and compared to the current state-of-the-art methods addressing the skin layer segmentation. The obtained Dice coefficient equal to 0.87 and 0.83 for the epidermis and SLEB, respectively, proves the developed framework's efficiency, outperforming the other approaches.

1. Introduction

During the last decades, inflammatory skin diseases have become an integral aspect of many people's life. They cover a broad range of disorders, including psoriasis, eczema, atopic dermatitis (AD), or acne, which are the most commonly recorded. Due to such symptoms like itching or dry skin, they lead to quality of life reduction [1]. Different methods reduce inflammatory skin disorders, including oral medications, steroid creams and ointments, phototherapy, and biologic treatment [2]. Monitoring of the treatment results and skin condition seem to be crucial for the therapy applied.

The fast development of imaging techniques, like high-frequency ultrasound (HFUS) or optical coherence tomography (OCT) makes it possible to monitor changes in skin layers reliably, accurately, and with a reduced time. HFUS is described as noninvasive, convenient, practical, and safe [3], therefore it is gaining popularity in dermatology. Nowadays, the ultrasounds at frequencies between 20 and 75 MHz are

commonly used in medical practice. The increasing ultrasound frequency reduces its penetration. Instead, the structures of the skin, including epidermis, dermis, subcutaneous fat layer, the muscle layer, blood vessels, hair follicles, and skin appendages, are clearly visible [3,4]. First, the treatment effect in atopic patients observed using HFUS (20 MHz) was described by Polanska et al. [5]. At the same time, Lin et al. [6] published the paper concerning the potential of using HFUS (18 MHz) in the early diagnosis of psoriasis arthritis. As reported in [5,7], the presence of a subepidermal low echogenic band (SLEB) below the echo entry in high-frequency ultrasound images indicates the presence of inflammatory dermatoses. It has been proven [5], that its thickness correlates with the severity of skin lesions in atopic dermatitis. Therefore, it can be used as an objective parameter in monitoring the course of the disease. To conclude, the changes of inflammatory skin condition may be visualized by changes of the SLEB thickness and increase in the intensity of the dermis image [5,7].

The fast development of HFUS imaging resulted in the development

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of related image processing techniques [7–11]. Despite various application possibilities, the US imaging is still not common in dermatology. Therefore, different methods are applied to various US frequencies (10 MHz [11] – 75 MHz [7,8]). The historically first paper [11] in the US-based skin analysis presents an automatic skin segmentation tool (ASST) to evaluate the skin toxicity in breast cancer radiotherapy. The developed framework introduces a geometric active contour model to skin segmentation and thickness measurement. The analyzed database includes 730 HFUS (10 MHz) images. The obtained segmentation masks compared to expert delineations resulted in a 95% accuracy. However, as reported in [9,10,7], the newest methods outperform the Gao et al. [11] framework. Furthermore, the 10 MHz is not sufficient for diagnosing inflammatory skin disease because the characteristic SLEB layer is not visible. Next, Sciolla et al. [9] applied a level set method to skin layers (epidermis and dermis) segmentation. Additionally, they segment the skin tumors visible in HFUS (> 20 MHz). This preliminary study in skin analysis was followed by their next paper [10], where the authors present novel joint segmentation and characterization algorithm for the assessment of skin aging using 50 MHz frequency US images. The described segmentation algorithm takes into account both the epidermis and SLEB layers. However, the SLEB segmentation results were not compared with the expert delineations. Moreover, the analyzed data set does not contain inflammatory patients. Our first work [12] in this area targets the epidermis layer segmentation in HFUS (22 MHz) images. This preliminary study, employing the fuzzy C-means clustering (FCM) followed by the fuzzy connectedness (FC), was continued in [7]. The presented framework is combining the level set algorithm [13] with a novel active contour model based on weighted fuzzy C-means (WFCM) [14] clustering results. Since deep learning-based methods are now firmly established as a robust tool, widely used in image segmentation [15–17], our latest work [8] present the preliminary results in this area. The paper describes the segmentation method, which employs a convolutional neural network (CNN) with a U-Net architecture to the SLEB segmentation in HFUS images of patients with AD. The promising results encourage us to redevelop this approach in the current study.

The deep learning-based approaches are able to process natural data, so they found applications in biomedical image segmentation [18–21,8]. The increased use of deep learning solutions is facilitated by computations involving graphical processing units (GPU), reducing the training and execution time [17]. Among many other AI solutions, CNN is the most widely used one. It consists of different layers responsible for specific operations, e.g., convolution, activation, pooling, loss calculation, etc. [15]. The dense pixel-wise prediction ability and improved segmentation results were possible by introducing the fully convolutional network (FCN) developed by Long et al. [22]. A deep convolutional encoder-decoder architecture – SegNet – was described by Badrinarayanan et al. [23] and successfully applied to the medical image dataset, e.g., by Khagi and Kwon [24].

Ronneberger et al. [18] proposed the most widely applied solution in medical image segmentation: the U-Net architecture incorporating the concept of deconvolution layers. Thanks to the superior design of skip connections between different stages, the network benefits in working with very few training images, yielding precise segmentation results. The advantage of using U-Net for HFUS image segmentation was presented in [8]. U-Net is able to train over a limited number of training samples, having a flexible and extensible structure [25]. Due to the similar appearance of OCT images – visible tissue layers, the most relevant work for our application was provided by Roy et al. [20]. The authors present the combination of U-Net and deconvolution net [26].

The skin layers were also segmented by Kepp et al. [27], who introduced densely-connected (DC) blocks as the modification of original U-Net architecture. In the newest approach, inspired by the Inception-ResNet structure [21], Gu et al. [19] proposed CE-Net architecture. The original U-Net architecture, with the limited scaling range, was extended by dense atrous convolution (DAC) block extracting enriched feature representations with multiscale atrous convolutions

[21]. The DAC block is followed by residual multi-kernel pooling (RMP) operations encoding the multiscale context features. As the authors claim [21], it outperforms the previous approach [20] in retinal OCT layer segmentation. Segmentation of epidermis regions and the follicular structures (which can be identified in the skin OCT images) is also described in [28]. The developed technique applies the U-Net architecture, followed by the Savitzky-Golay filter and Fourier domain filtering steps.

On the other hand, the well known unsupervised clustering approaches, including the FCM, are still widely used in medical image segmentation, standing alone [29] or improving the CNN results [30,31]. In the first of the combined approaches [30], the FCM-based segmentation is used for textural feature extraction and then followed by CNN. The developed framework was tested on lung computed tomography (CT) image data. The newest technique [31], addressing the brain tumor segmentation problem in the magnetic resonance (MR) images, uses the FCM clustering for the region of interest (ROI) detection. The final segmentation step is based on the CNN architecture, again.

The latest research in HFUS image analysis [7,10] reported that the accurate and reliable segmentation of skin layers is useful in monitoring inflammatory dermatoses. To meet this need, we introduce an algorithm for automated skin layers segmentation (epidermis and SLEB) in HFUS images (75 MHz) in patients with inflammatory skin diseases, including psoriasis and AD. To the best of our knowledge, the problem of skin layers segmentation in HFUS images using deep learning methods is addressed only in [8]. The U-Net segmentation results of SLEB are promising but preliminary due to the limited data set, including 45 HFUS images of patients with AD. In our current work, we present the CNN based segmentation technique verified on the database consisting of 380 HFUS images of patients with both AD and psoriasis. We continue our previous research [8] by introducing the combination of FCM clustering results with different CNN architectures. We show that the three-channel input combining gray image, Lab color space, and FCM clustering results improves the CNN based segmentation algorithm operating on the original RGB data. For this, both the original U-Net and the CE-Net [19] architecture (successfully applied to the retina layer segmentation) are considered. Besides, we propose a novel SegUNet framework, including batch normalization layers, which adjust and scale the activation, making the segmentation results more stable. Finally, we compare the obtained results with the current state-of-the-art study in skin layer segmentation in HFUS images [7]. The obtained results demonstrate that the presented workflow successfully segment the target layers and outperforms the state-of-the-art methods [7,8,10] in this area. We share the trained models along with the complete database and expert delineations through Mendeley Data.

The organization of the manuscript is the following. Section 2 describes the developed segmentation methodology, including three steps: preprocessing, CNN based segmentation algorithm, and postprocessing. The image database, performed experiments, and the obtained results confirming the advantages of our approach are given in 3. The paper ends with a discussion and conclusions in Sections 4 and 5, respectively.

2. Materials and methods

The paper presents an automatic algorithm able to segment skin layers: epidermis and SLEB in HFUS images of patients with inflammatory skin diseases: AD and psoriasis. The developed method combines a similarity-based clustering algorithm (FCM) with a fully convolutional network and robust postprocessing step. The processing framework from the input image to the final segmentation mask is shown in Fig. 1.

2.1. Preprocessing

A two-step preprocessing method is introduced to increase the accuracy of the developed segmentation algorithm. First, considering the

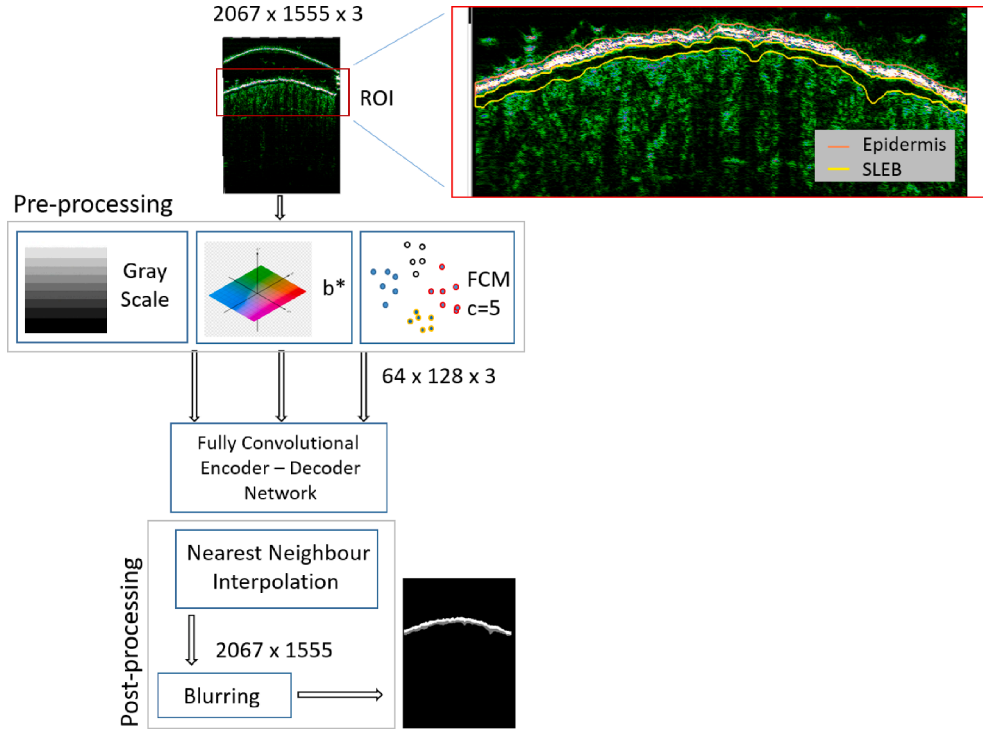


Fig. 1. Block diagram of the processing framework.

analysis of quite small data set for CNN training and according to the conclusions from [8,28], the analyzed image was limited to the region of interest (ROI) surrounding the epidermis layer. It reduces false positive segmentations in the US probe area above the epidermis and US probe shadow below the SLEB layer. For this, we apply the epidermis layer detection algorithm described in [7]. Let the set of (x, y) represents the coordinates of the detected epidermis, and I stands for the original RGB input image of size $N_x \times N_y$. The ROI is limited along the x direction, resulting in I_{ROI} . To ensure, that the ROI contains all parts of interest, we define its size based on the statistical skin layer thickness [32], so that the new coordinates are given as (x_{ROI}, y) , where: $x_{ROI} \in [\max(1, \min(y) - 0.5mm), \min(N_x, \max(y) + 1mm)]$.

The next preprocessing step replaces the original RGB channels of the input image I_{ROI} by the combination of a grayscale image, CIE $L^*a^*b^*$ color space [33], and weighted fuzzy C-means [14] clustering results. Since the blue channel is described as the most informative for the skin layer segmentation [7], the b^* channel expressing the color values from blue to yellow is selected for further analysis. Due to the benefits of combining FCM clustering with the CNN architecture [30,31], the WFCM pre-segmentation step is used to create one of the input channels of the image I_{ROI} . The WFCM is a more efficient version of the FCM, which speeds up the original analysis, replacing the original data by its histogram. We set the desired number of clusters to five, representing all the skin layers, US gel (background), and the subcutaneous fat. In the WFCM layer of the image I_{ROI} , the brightest area (epidermis) is represented by the brightest cluster. Finally, the input data are rescaled to the size of $64 \times 128 \times 3$.

2.2. CNN architecture

The skin layer segmentation is defined as the classification problem of three clusters, since the three areas under consideration are: epidermis, SLEB, and the remaining image part. Our encoder-decoder architecture (SegUNet) is shown in Fig. 2. Four contraction layers consist of a pair of, in sequence: convolutional layer, batch normalization layer, ReLU layer followed by max-pooling layer. The batch normalization layer, adopted from SegNet [23] architecture, adjusts and

scales the activation, makes the segmentation results more stable, and increases the neural network performance. The max-pooling layer of a pool size 2×2 and stride 2 reduces the spatial information. Each convolutional layer is composed of N filters of size 3×3 , where N starts from 64 and is doubled in each layer, reaching a maximum of 1024 in an expansion section.

Each encoder block is connected with the corresponding decoder block by the concatenation layer. The decoder blocks consist, again, of a pair of, in sequence: convolutional layer, batch normalization layer, and ReLU layer, applied to the concatenated feature map. The last convolutional layer is followed by a softmax activation function and segmentation layer, which associate each image pixel to the desired class.

The encoder and decoder paths are connected by the expansion section [18], which is similar to the contraction layer. It consists of a dropout layer, a max-pooling layer, and two convolutional layers followed by ReLU layers.

We trained the network by optimizing the cross-entropy loss through the stochastic gradient descent with momentum (SGDM) optimizer [34], with the momentum set to 0.9. To reduce overfitting, a regularization term is added to the loss function $E(\theta)$ [34], yielding a loss function formula:

$$E_R(\theta) = E(\theta) + \lambda \Omega(\mathbf{w}), \quad (1)$$

where θ is the network parameter vector (weights and biases), $\mathbf{w}$ stands for the weight vector, $\lambda = 0.0005$ is the regularization coefficient, and $\Omega(\mathbf{w})$ represents the regularization function:

$$\Omega(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w}. \quad (2)$$

Each training process consists of 200 epochs. The training data are shuffled before each training epoch. The batch size is 8, and the initial learning rate is equal to 0.001. The training data are augmented by introducing random geometric transformations (reflection in the horizontal direction, ± 10 -pixel translation in both directions, $\pm 10^\circ$ rotation).

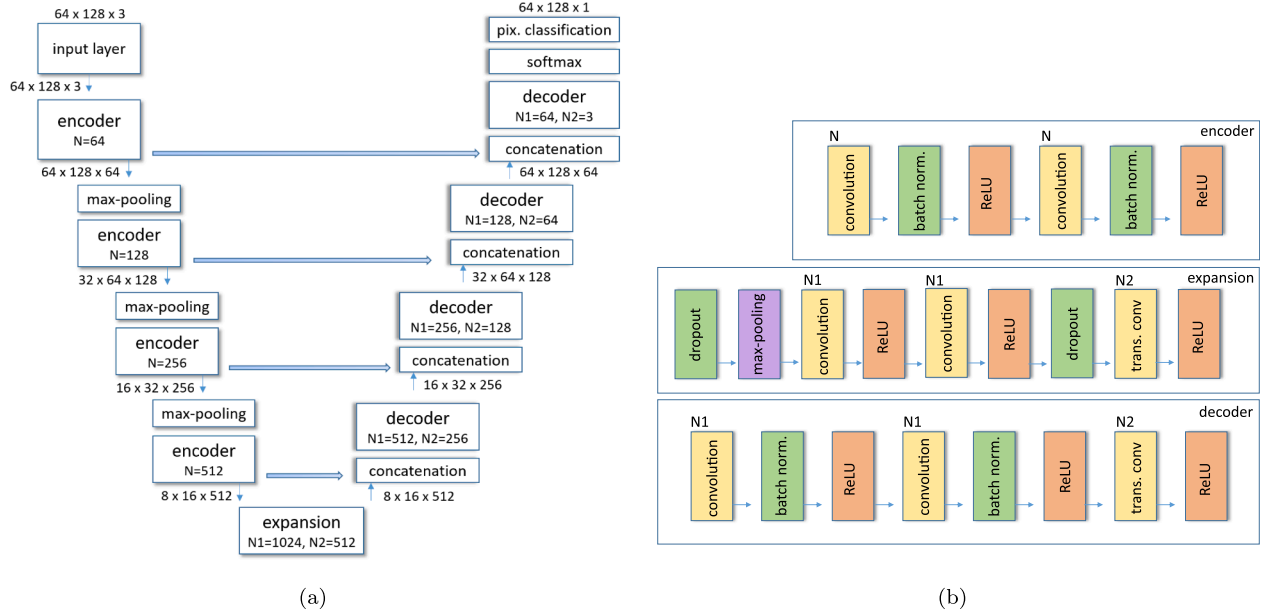


Fig. 2. SegUNet architecture (a), network blocks (b).

2.3. Postprocessing

The postprocessing procedure aims to reduce the number of false-positive detections and the influence of image interpolation (nearest-neighbor) to the final segmentation results. First, the morphological opening is applied to remove the small objects appearing in the US probe or US probe shadow areas, incorrectly classified as the epidermis regions. Second, assuming that the epidermis layer segmentation results are more accurate, and the SLEB layer is always located directly under the epidermis part, a one-pixel layer below the epidermis completes the SLEB segmentation results. This step reduces false-negative SLEB detections. Finally, to reduce the influence of image resizing, which results in sharp edges of the segmented regions, a blurring filter is applied, with a 5×5 kernel and a threshold equal to 0.5.

The obtained segmentation results are used to estimate the skin layer thickness. The layer thickness for both the expert delineation and segmentation results is determined as the mean distance, in millimeters, between the upper and lower layer boundary calculated along the layer area.

3. Experiments and results

The analyzed dataset consists of 380 HFUS images of 380 different patients with AD (303) and psoriasis (77). The data were acquired using DUB SkinScanner75 with a 75 MHz transducer yielding images of size 2067×1555 pix and of four different resolutions (lateral $\times$ axial): 0.0019×0.085 , 0.0024×0.085 , 0.0031×0.085 , 0.0019×0.085 mm/pix. In each case, the epidermis and SLEB were delineated by one expert and then verified by another expert, both working with HFUS images in their daily dermatological and scientific practice for four years (graduated from the Sono-Derm training course). To share our materials and methods with the scientific society, we have uploaded the complete database with annotations and trained models to Mendeley Data¹.

To compare the proposed framework with existing skin segmentation approaches, we analyze four different network architectures: CE-Net [19], basic U-Net [18], SegNet (vgg16) [23], and the proposed one (SegUNet). Moreover, the obtained results are compared with those

provided by the current state-of-the-art technique for the skin layer segmentation in HFUS images (FCM-AC) [7].

To justify using the combination of gray, L^*a^*b , and WFCM clustering results as the network input, we feed the deep networks with both input data representations: original RGB image (v0) and the image prepared according to our proposition (v1). In each experiment, we compare the segmentation results obtained within both v0 and v1 frameworks.

To avoid biased results and the influence of partitioning the data set into training and testing subsets, we use an external K-fold cross-validation technique. For this purpose, $K = 10$ partitions are created, and the remaining data are additionally divided into training and validation part, in the proportion of 8:1.

For the quantitative analysis of the obtained segmentation results, we use the pixel-wise Dice coefficient:

$$D = \frac{2TP}{2TP + FP + FN}, \quad (3)$$

where TP, FP, FN denote the number of true positive, false positive, and false negative pixels for a given class (epidermis or SLEB), respectively. Fig. 3 presents the results for the epidermis segmentation in both input data modes. Median Dice index values over ten aggregated testing sets are given in the charts along with the corresponding boxplots. The SLEB segmentation efficiencies are presented in Fig. 4 in the same way. Our approach performs better than the other methods in each case. On the other hand, the employment of the proposed input image (v1) generally yields more efficient segmentation compared to the analysis based on the original RGB image (v0). Only SegNet is a clear exception here.

Moreover, due to the fact, that the primary goal of the skin layer segmentation is the layer thickness estimation, we provide details of thickness measurements in both epidermis (d_{epid}) and SLEB (d_{SLEB}) layers. With the ground-truth thickness d_l^{gt} of a skin layer l ($l = epid$ or $SLEB$) obtained from the expert delineation, we can determine the layer thickness estimation error Δd_l produced by a given method:

$$\Delta d_l = d_l - d_l^{gt}, \quad (4)$$

where d_l is the thickness of the layer l given by a given method. Figs. 5 and 6 present the boxplots of the layer thickness estimation errors obtained by using all five methods under consideration for the epidermis and SLEB, respectively. The median error of the proposed approach does

¹ <https://data.mendeley.com/datasets/5p7fxjt7vs/draft?a=fe8f558b-4aa1-4753-a366-8a5e3bfa7ec8>.

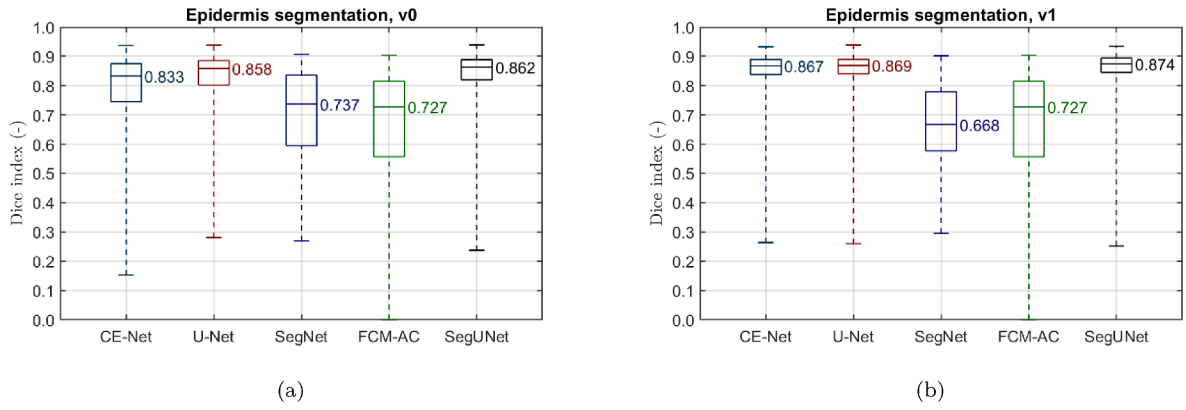


Fig. 3. Dice index box plots for the epidermis segmentation using our method (SegUNet) and four reference methods in the v0 (a) and v1 (b) mode. Each box covers a 25th to 75th percentile range with median value given and indicated by a central line. The extreme values are bordered by whiskers.

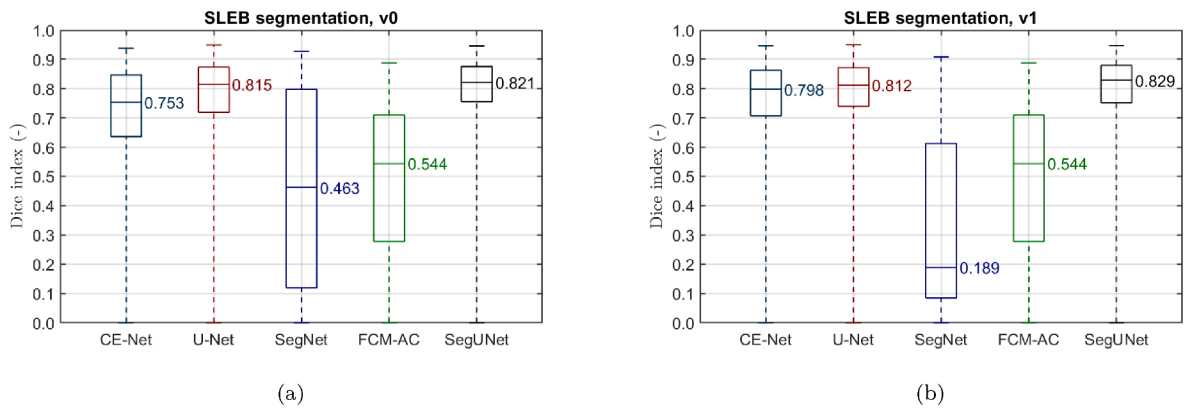


Fig. 4. Dice index box plots for the SLEB segmentation using our method (SegUNet) and four reference methods in the v0 (a) and v1 (b) mode. Each box covers a 25th to 75th percentile range with median value given and indicated by a central line. The extreme values are bordered by whiskers.

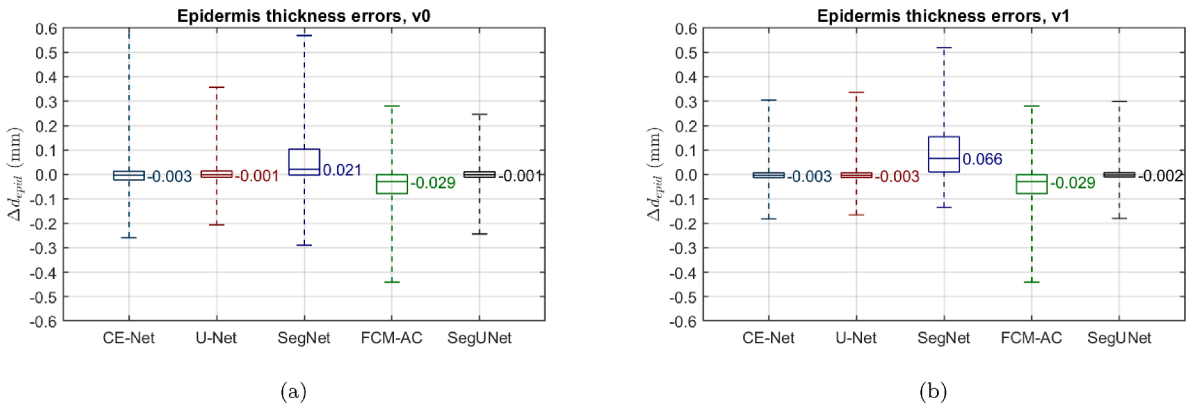


Fig. 5. Layer thickness error Δd_{epid} box plots for the epidermis segmentation using our method (SegUNet) and four reference methods in the v0 (a) and v1 (b) mode. Each box covers a 25th to 75th percentile range with median value given and indicated by a central line. The extreme values are bordered by whiskers.

not exceed ± 0.002 mm in any case and is never worse than any other method. Also, the dispersion of errors is the smallest of all methods.

For both layers, we also calculated the root mean square errors (RMSE) and the mean absolute errors (MAE) throughout all 380 cases. The results are presented in Table 1 for the epidermis and in Table 2 for the SLEB. Table 3 shows the models' size in terms of the number of learnable weights. The segmentation speed of each network, in frames per second (fps), is also presented in Table 3.

To assess the obtained results, we employed statistical analysis. First, we verified the normality of distributions of all groups of obtained skin layer thickness measures (five methods $\times$ two layers $\times$ two variants plus two ground truth sets for either layer) by using the Shapiro-Wilk test. Neither distribution was confirmed normal. Therefore, we employed the Wilcoxon matched-pairs signed-rank test to any pair of methods and each method vs. corresponding ground truth thicknesses. In each case, the null hypothesis states that the difference between two paired data-

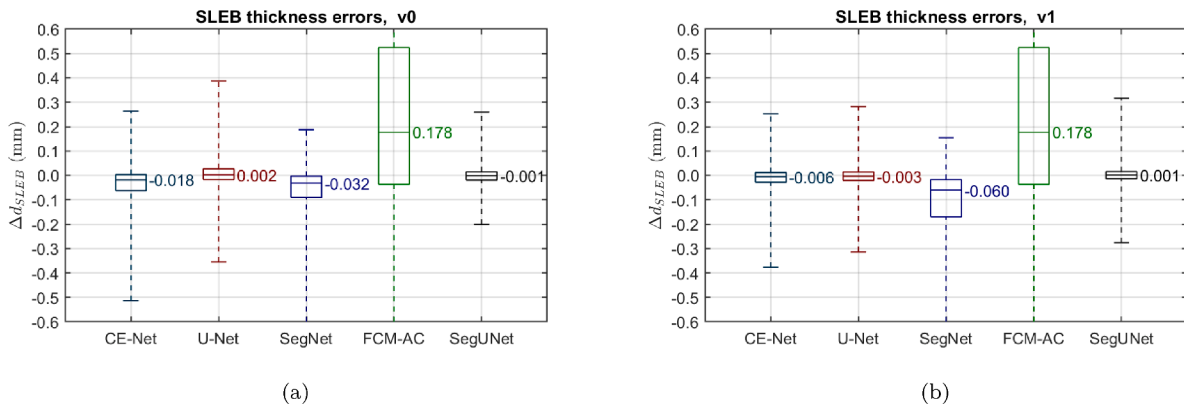


Fig. 6. Layer thickness error Δd_{SLEB} box plots for the SLEB segmentation using our method (SegUNet) and four reference methods in the v0 (a) and v1 (b) mode. Each box covers a 25th to 75th percentile range with median value given and indicated by a central line. The extreme values are bordered by whiskers.

Table 1

Summary of the epidermis layer thickness errors (MAE and RMSE) produced by our method (SegUNet) and four reference methods in both v0 and v1 modes. All values in millimeters. Minimum errors indicated with a bold font.

		CE-Net	U-Net	SegNet	FCM-AC	SegUNet
MAE	v0	0.040	0.025	0.080	0.415	0.021
	v1	0.015	0.015	0.108	0.415	0.013
RMSE	v0	0.093	0.048	0.133	0.665	0.037
	v1	0.027	0.027	0.152	0.665	0.025

Table 2

Summary of the SLEB layer thickness errors (MAE and RMSE) produced by our method (SegUNet) and four reference methods in both v0 and v1 modes. All values in millimeters. Minimum errors indicated with a bold font.

		CE-Net	U-Net	SegNet	FCM-AC	SegUNet
MAE	v0	0.058	0.040	0.083	0.113	0.029
	v1	0.041	0.034	0.116	0.113	0.029
RMSE	v0	0.093	0.069	0.137	0.159	0.045
	v1	0.071	0.057	0.168	0.159	0.051

Table 3

Summary of the deep model's number of trainable parameters and time consumption (v1 mode). Experiments performed in Matlab R2020b on a workstation with an 8-Core CPU @ 3.20 GHz, 64 GB RAM, and Nvidia Quadro RTX 6000 24 GB GPU.

	CE-Net	U-Net	SegNet	SegUNet
Number of trainable parameters	123,329,103	31,031,875	29,444,745	31,039,555
Segmentation speed (fps)	83.45	89.91	79.86	88.42

Table 4

Results of the pairwise Wilcoxon signed-rank test (p -values) for the skin layer (epidermis and SLEB) thickness produced by different methods in both v0 and v1 modes vs. the corresponding ground truth measures. Values of $p \geq .05$ (no statistical difference between the method's outcome and the ground truth measures) indicated with a bold font.

		CE-Net	U-Net	SegNet	FCM-AC	SegUNet
Epidermis	v0	.04	.31	<.001	<.001	.97
	v1	.001	.002	<.001	<.001	.07
SLEB	v0	<.001	.052	<.001	<.001	.19
	v1	<.001	.04	<.001	<.001	.70

sets comes from a zero median distribution. In the method-to-method comparisons, the null hypothesis was confirmed ($p \geq .05$) in only a couple of cases, both for the epidermis: U-Net vs. SegUNet in the v0 mode and CE-Net vs. U-Net in the v1 mode. In all other cases illustrated in Figs. 5 and 6, the results were recognized statistically different. Table 4 presents the p -values yielded by the Wilcoxon test for each method-to-ground-truth comparison. Our SegUNet approach produces layer thickness statistically not different than the ground truth thickness in all four cases. From other methods, only the U-Net matched the ground truth measures ($p \geq .05$) in the v0 mode.

In Figs. 7 and 8, we present two examples of segmentation outcomes produced by our method and two other methods: CE-Net and U-Net, with the lowest thickness errors. The visual improvement between the two proposed modes of the analysis is especially visible in CE-Net. The segmentation results of all three architectures are acceptable.

4. Discussion

In this work, we propose a novel framework for the segmentation of skin layers (epidermis and SLEB) in HFUS images. The analyzed data set consists of 380 images of patients with AD and psoriasis in unequal proportion (303 to 77). The dataset includes manual delineation of both epidermis and SLEB layers, performed by an expert. The obtained results are compared to expert delineations and previous studies in this area [7,8]. The FCM-AC technique [7], based on the image data from 45 patients, introduces the novel active contour approach and results in the Dice coefficient of 0.87 for the SLEB segmentation. It is worth to underline that only one previous work [8] applies the CNN based approach for HFUS skin image segmentation. However, due to the limited database, the results described there are rather preliminary. Despite this, they are comparable to [7] for the same database of 45 cases. Since the CNN segmentation techniques are successfully applied in medical image analysis, we decided to find the application domain with the most comparable appearance, and use the solutions described there to our segmentation task. Therefore, we involved two network architectures [19,28], initially developed for the OCT image segmentation. The OCT image datasets described in [19,28] include retinal and skin images with tissues of a characteristic layered structure. The first method – the CE-Net [19] – yields a mean absolute error of 1.68 pixel for the retina OCT database (20 3D volumes, 256 images each, 11 delineated layers). The second one – the U-Net [28] – produces the Dice coefficient of 0.83 for the epidermis segmentation in the OCT database consisting of 270 skin images. The most interesting and novel part of this work is the postprocessing step based on filtering procedures, which reduces false positive detection of hair follicles, not applicable in current work. The last architecture under consideration is the SegNet (vgg16) [23] successfully applied for medical image analysis in the past, still presenting a

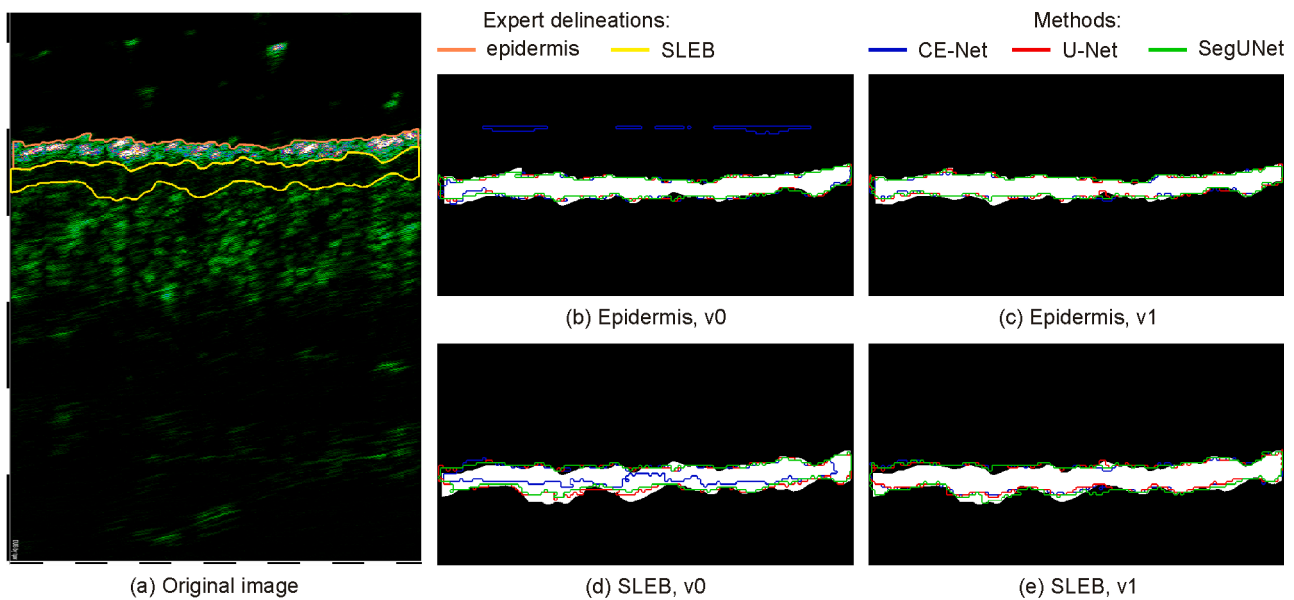


Fig. 7. Sample HFUS image (case #349) with expert delineations of the epidermis (orange) and SLEB (yellow) regions (a). Contours of segmentation outcomes from three methods: CE-Net (blue), U-Net (red), and SegUNet (green) in two modes v0 and v1 superimposed on the binary objects from the expert delineations of the epidermis (b), (c) and SLEB (d), (e).

potential for various applications.

The first remark from the performed numerical analysis is that the newest solutions, based on CNN, applied for the first time to such an extensive and representative dataset of skin HFUS images outperform the active contour technique [7]. Despite the active contour methods are described as robust and accurate in medical image segmentation, different constraints limit their application area. For the small dataset of similar cases [7], the active contour approach results in a high Dice coefficient of 0.87 for the SLEB segmentation. However, the same algorithm applied to a more varied dataset, including different skin diseases, the obtained results are less impressive and fall to 0.54 and 0.73 for the SLEB and epidermis segmentation, respectively. For the same dataset, two of the applied CNNs (U-Net, novel SegUNet) result in a Dice

coefficient greater than 0.8 for both analyzed layers (see Figs. 3 and 4). The same conclusions can be drawn based on the estimated RMSE and MAE of the skin thickness measurements (see Tables 1 and 2).

From the experiment analyzing the influence of the proposed pre-processing step (v1 mode) on the obtained CNN segmentation results, we can conclude, that it improves segmentation results in most U-shaped architectures. It is especially noticeable in epidermis layer segmentation (see Fig. 3 and Table 1), where the Dice coefficient increases after incorporating this step. The difference in the obtained segmentation results are also apparent in Figs. 7 and 8. This improvement is not clearly visible in SegNet (vgg16). That can be explained by the network architecture, which does not capture multiscale information so effectively. Moreover, the pre-trained weights of SegNet are used at the

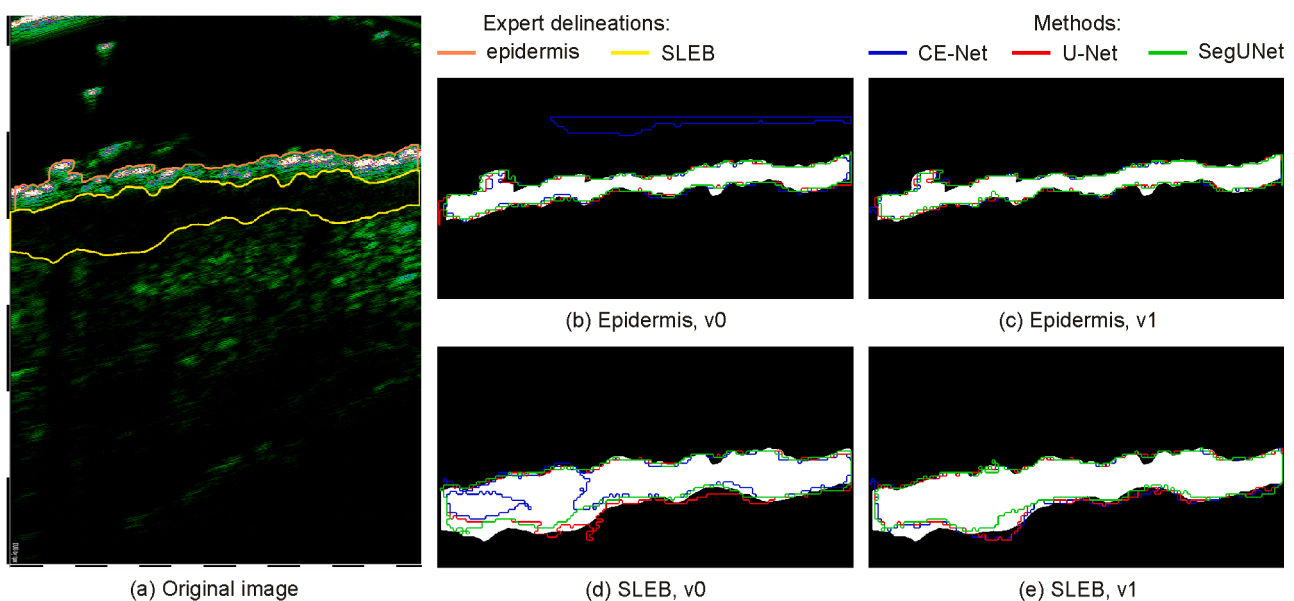


Fig. 8. Sample HFUS image (case #351) with expert delineations of the epidermis (orange) and SLEB (yellow) regions (a). Contours of segmentation outcomes from three methods: CE-Net (blue), U-Net (red), and SegUNet (green) in two modes v0 and v1 superimposed on the binary objects from the expert delineations of the epidermis (b), (c) and SLEB (d), (e).

initialization stage, while the original model was trained using RGB images.

All the carried experiments prove the benefits of using the proposed framework in HFUS skin layer segmentation and measurements compared to existing approaches in the layered image segmentation [19,28]. However, based on the statistical analysis, we can conclude that the U-Net also produces thickness measurements comparable to the expert (see Table 4). Finally, based on thickness measurement errors, it is noticeable that the results provided by the proposed model are more stable. The dispersion of layer thickness errors produced by the SegUNet is the smallest (see Figs. 5 and 6).

The computer-aided skin diagnosis may seem underestimated thus far, mainly due to the HFUS application in this domain being in its early stages. However, we can expect an increasing biomedical engineers' interest here in a short time. Our study introduces deep learning with additional image processing to address the epidermis and SLEB segmentation more efficiently than the state-of-the-art approaches. Our model can be highly applicable for HFUS skin diagnosis with an ability to operate in real-time (note Table 3). Moreover, we share our database and models publicly to enable future research in the domain.

5. Conclusion

The paper presents a novel CNN-based framework for the HFUS image segmentation. The developed approach makes it possible to segment both epidermis and SLEB layers with comparable accuracy (0.87 and 0.83). It is the first CNN-based approach that addresses the problem of analyzing such a big data set of real HFUS images of patients with AD and psoriasis. Based on the obtained results, it is possible to conclude that the CNN-based methods outperform the classical segmentation techniques [7]. Moreover, the thickness measurements performed based on the areas segmented by our method are comparable to the manual expert delineations. Finally, the proposed algorithm outperforms the existing approaches in HFUS image segmentation and the newest segmentation techniques in OCT image analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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